METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

Chulalongkorn University · Department of Chemical Engineering

DATASHEET E-204

E-204 · Start-up heater (HP steam)

DOCUMENT NO.	:	MEOH-V1-ME-DS-006
DOCUMENT TITLE	:	Datasheet E-204
PROJECT	:	Methanol Converter Ver 1 · OTS
STATUS	:	Rev 0 · Issued for teaching

Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

E-204 · Start-up heater (HP steam)

Equipment list MEOH-V1-ME-EQL-001. Values from the simulator model at its design point unless marked assumed. P&ID: MEOH-PID-001.

General		
Service	Start-up heater (HP steam)	MEOH-PID-001
Type	BEU: process in U-tubes, condensing HP steam in the shell; horizontal	TEMA (assumed)
Shells	1 (single)	
Performance (model, design point)		
Duty	22.2 MW	start-up rating case: loop gas 150 °C in, HV-214 open (assumed)
Hot side in → out	252 → 252 °C	
Cold side in → out	150 → 214.5 °C	
LMTD / F	64.5 K / 1	condensing steam on the shell side: F = 1
Model UA	345 kW/K	K.NTUh × 345 kW/K (model at full opening)
Service U (with fouling)	500 W/m²K	assumed, typical
Installed area	760 m²	model UA + 10 % margin (assumed)
Design		
Tube side (loop gas)	100 bar g / 335 °C	Zone B
Shell side (HP steam 40 bar g)	45 bar g / 280 °C	assumed: sat. 252 °C + margin
Maximum duty	30 MW (K.QHmax)	HV-214 limit in the model
Code	ASME VIII Div. 1 (assumed); TEMA R for exchangers	
Corrosion allowance	3 mm (CS / Cr-Mo parts), 0 (tubes)	assumed
Construction		
Tubes	2082 × OD 19.05 × 2.11 mm × 6.1 m	triangular pitch 23.8 mm
Shell inside diameter	1250 mm	tube count by TEMA bundle factors
Materials	1¼Cr-½Mo SA-387 Gr 11 Cl 2 channel and tubes (process); shell SA-516 Gr 70 (steam)	EQL §3
Thickness (pressure side)	channel 44.5 + 3 → 48 mm (ID 1250)	UG-27(c)(1)
Notes		
1	In service only at start-up (HIC-214); HV-214 blocked in at design.	MEOH-PID-001 note 5
2	Tube rupture: loop gas into the 45 bar g steam side — relief required on the steam side (RLB, EQ-er).	